```
➜  04 python3 generator.py data.xml schema.xsd
[SUCCESS] Generated XML Schema: schema.xsd
[INFO] Root element: students
[INFO] Output file: /Users/linyh/Documents/Grade04/XML/HW/04/schema.xsd
➜  04 cat schema.xsd
<?xml version='1.0' encoding='UTF-8'?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="students">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="student" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="name" type="xs:string"/>
              <xs:element name="email" type="xs:string"/>
              <xs:element name="age" type="xs:integer"/>
              <xs:element name="department" type="xs:string"/>
              <xs:element name="courses">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="course" type="xs:string" maxOccurs="unbounded"/>
                  </xs:sequence>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
            <xs:attribute name="id" type="xs:ID" use="required"/>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
➜  04 python3 validator.py invalid_data.xml schema.xsd
[INVALID] invalid_data.xml does not conform to schema.xsd.
  Line 3, column 0, Element 'student', attribute 'status': The attribute 'status' is not allowed.
  Line 6, column 0, Element 'department': This element is not expected. Expected is ( age ).
  Line 22, column 0, Element 'department': This element is not expected. Expected is ( age ).
```

Results:

1. Schema Generation

- **Command**: python3 generator.py data.xml schema.xsd
- **Description**: Parses data.xml, identifies the root element <students>, infers element types/structures, and successfully writes schema.xsd.

2. Successful Validation

- **Command**: python3 validator.py data.xml schema.xsd
- **Description**: Confirms data.xml conforms to the generated schema.xsd with [SUCCESS].

3. Failed Validation (Error Handling)

- **Command**: python3 validator.py invalid_data.xml schema.xsd
- **Description**: Tests error detection on invalid_data.xml. The validator successfully catches and logs two schema violations:
    1. An undeclared status attribute on <student> at Line 3.
    2. Element sequence mismatch (<department> placed before <age>) at Lines 6 and 22.